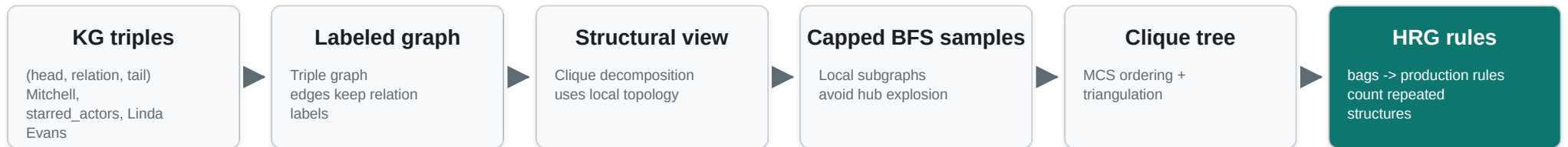
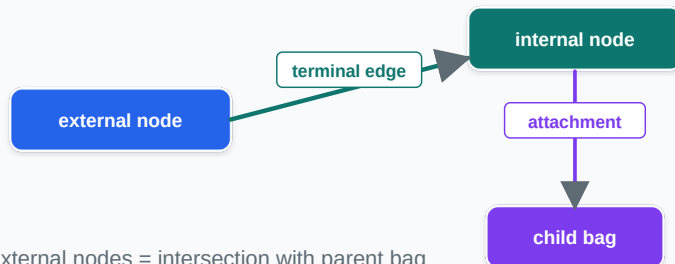


Offline HRG grammar extraction

This figure explains how structural grammar rules are learned from the KG before question answering.



How one bag becomes a rule



external nodes = intersection with parent bag
internal nodes = new nodes introduced in this bag
terminal edges = KG edges assigned to this bag

Resulting production rule

LHS: N / rank

RHS:

terminal_edges = [(a, relation, b), ...]

nonterminals = [N / rank with attachments]

count = how often this structure appears

Use this as a structural prior, not as a full grammar decoder.